WoSM Academy

MAT231-F27 · CALCULUS II

Partial derivatives and differentiability

Partial derivatives; differentiability; the total derivative

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Last time

- Norms on $\mathbb{R}^n$; all of them give the same limits.
- Open sets, closed sets, compact (closed and bounded) sets.
- Limits must agree along every path: $\frac{xy}{x^2+y^2}$ has no limit at $(0,0)$, and $\frac{x^2y}{x^4+y^2}$ fools every straight line.
- Continuous in each variable separately does not mean continuous.

Today: derivatives in $\mathbb{R}^n$. The obvious idea — differentiate in one variable at a time — is useful, but not enough. The right notion is a *linear approximation*.

This lecture

01 Partial derivatives

02 Partial derivatives are not enough

03 The total derivative

04 The Jacobian matrix

05 Summary and practice

Reading: G. Strang, E. Herman, *Calculus, Volume 3* (OpenStax), Sections 4.3 and 4.4.

01

Partial derivatives

Freeze all variables but one

Definition

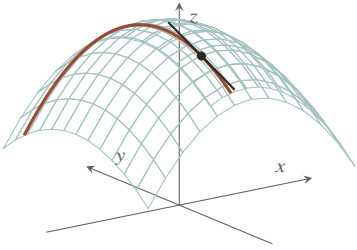
Let $f : U \to \mathbb{R}$, $U \subseteq \mathbb{R}^n$ open, $\mathbf{a} \in U$, and $\mathbf{e}_i$ the i-th unit vector. The **partial derivative** of f with respect to x_i at $\mathbf{a}$ is

$$\frac{\partial f}{\partial x_i}(\mathbf{a}) = \lim_{t \to 0} \frac{f(\mathbf{a} + t\,\mathbf{e}_i) - f(\mathbf{a})}{t},$$

if the limit exists. Other notations: $f_{x_i}(\mathbf{a})$, $\partial_i f(\mathbf{a})$, $D_i f(\mathbf{a})$.

For $f(x, y)$: $f_x(a, b)$ is the ordinary derivative of the one-variable function $x \mapsto f(x, b)$ at $x = a$. All other variables are constants.

A picture



$f(x, y) = 4 - x^2 - \frac{1}{2}y^2$.

- Cut the surface with the plane $y = 1$: the curve $z = 3.5 - x^2$.
- $f_x(1, 1)$ is the slope of this curve at $x = 1$: -2.
- Likewise $f_y(1, 1) = -1$ is the slope along the cut $x = 1$.

Partial derivatives see only the directions of the coordinate axes.

Computing partial derivatives

Use the rules of one-variable calculus, treating the other variables as constants.

- $f(x, y) = x^2 y + \sin(xy)$: $f_x = 2xy + y\cos(xy)$, $\quad f_y = x^2 + x\cos(xy)$.
- $f(x, y, z) = x\, e^{yz}$: $f_x = e^{yz}$, $\quad f_y = xz\, e^{yz}$, $\quad f_z = xy\, e^{yz}$.
- $f(\mathbf{x}) = \|\mathbf{x}\| = \sqrt{x_1^2 + \cdots + x_n^2}$, $\mathbf{x} \neq \mathbf{0}$: $\dfrac{\partial f}{\partial x_i} = \dfrac{2x_i}{2\sqrt{x_1^2 + \cdots + x_n^2}} = \dfrac{x_i}{\|\mathbf{x}\|}$.

A function of many weights

For a loss $L(w_1, \ldots, w_n)$, $\frac{\partial L}{\partial w_i}$ says how much the loss changes when only the weight w_i moves. Training a model needs all n of them at once (Units 11 and 15).

Higher partial derivatives

Differentiate again: $f_{xx} = (f_x)_x$, $f_{xy} = (f_x)_y$, f_{yx}, f_{yy}; also written $\frac{\partial^2 f}{\partial x^2}$, $\frac{\partial^2 f}{\partial y\,\partial x}$, …

Example

$f = x^2 y + \sin(xy)$: $f_{xy} = 2x + \cos(xy) - xy\sin(xy) = f_{yx}$.

Theorem (Schwarz, or Clairaut)

If f_{xy} and f_{yx} exist near $\mathbf{a}$ and are continuous at $\mathbf{a}$, then $f_{xy}(\mathbf{a}) = f_{yx}(\mathbf{a})$.

Without continuity it can fail: for $f = \frac{xy(x^2-y^2)}{x^2+y^2}$, $f(0,0) = 0$, one finds $f_{xy}(0,0) = -1$ but $f_{yx}(0,0) = 1$. The Hessian (Unit 13) is symmetric thanks to this theorem.

02

Partial derivatives are not enough

A warning from Lecture 9

Example

$f(x,y) = \frac{xy}{x^2+y^2}$ for $(x,y) \neq (0,0)$, $f(0,0) = 0$.

- $f(x,0) = 0$ and $f(0,y) = 0$, so $f_x(0,0) = 0$ and $f_y(0,0) = 0$.
- But f is not even continuous at $(0,0)$.

- In one variable, a differentiable function is continuous. So "all partial derivatives exist" cannot be the right meaning of "differentiable" in $\mathbb{R}^n$.
- The partial derivatives only look along two lines through the point. A good derivative must describe f near **a** in *all* directions at once.

03

The total derivative

Back to one variable

$f\colon \mathbb{R} \to \mathbb{R}$ is differentiable at a if and only if there is a number $f'(a)$ with

$$f(a+h) = f(a) + f'(a)\,h + r(h), \qquad \frac{r(h)}{h} \to 0.$$

- $h \mapsto f'(a)\,h$ is a *linear map* $\mathbb{R} \to \mathbb{R}$: the best linear approximation of $f(a+h) - f(a)$.
- The error $r(h)$ is small *compared with* h: we write $r(h) = o(h)$.
- This version generalises: replace the number $f'(a)$ by a linear map $\mathbb{R}^n \to \mathbb{R}^m$, that is, by a matrix.

Differentiability in $\mathbb{R}^n$

Definition

Let $U \subseteq \mathbb{R}^n$ be open. $f : U \to \mathbb{R}^m$ is **differentiable at $\mathbf{a} \in U$** if there is a linear map $L : \mathbb{R}^n \to \mathbb{R}^m$ (an $m \times n$ matrix) such that

$$\lim_{\mathbf{h} \to \mathbf{0}} \frac{\| f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a}) - L\mathbf{h} \|}{\|\mathbf{h}\|} = 0.$$

L is the **total derivative** of f at $\mathbf{a}$, written $Df(\mathbf{a})$.

- In words: $f(\mathbf{a}+\mathbf{h}) = f(\mathbf{a}) + Df(\mathbf{a})\,\mathbf{h} + o(\|\mathbf{h}\|)$.
- $\mathbf{h} \to \mathbf{0}$ in every way at once — this is where the paths of Lecture 9 come in.
- f is differentiable on U if it is differentiable at every point of U.

What the derivative must be

Theorem

If f is differentiable at $\mathbf{a}$, then

1. f is continuous at $\mathbf{a}$;
2. all partial derivatives $\partial f_i/\partial x_j\,(\mathbf{a})$ exist, and $Df(\mathbf{a})$ is the $m \times n$ matrix of them: row i, column j holds $\partial f_i/\partial x_j\,(\mathbf{a})$.

Proof.

1. $f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a}) = L\mathbf{h} + o(\|\mathbf{h}\|) \to \mathbf{0}$. 2. With $\mathbf{h} = t\,\mathbf{e}_j$: $\frac{f(\mathbf{a}+t\mathbf{e}_j)-f(\mathbf{a})}{t} \to L\mathbf{e}_j$, the j-th column of L; so that column is $\partial f/\partial x_j\,(\mathbf{a})$. □

So $Df(\mathbf{a})$ is unique, and we compute it with partial derivatives — once it exists.

The case $m = 1$: tangent planes

For $f\colon \mathbb{R}^2 \to \mathbb{R}$, $Df(a,b) = \begin{pmatrix} f_x(a,b) & f_y(a,b) \end{pmatrix}$, a 1×2 matrix, and

$$f(a+h, b+k) \approx f(a,b) + f_x(a,b)\,h + f_y(a,b)\,k.$$

The graph of the right-hand side is the **tangent plane** $z = f(a,b) + f_x(a,b)(x-a) + f_y(a,b)(y-b)$.

> **Example ($f(x,y) = x^2 + y^2$ at $(1,2)$)**
>
> $f = 5$, $f_x = 2$, $f_y = 4$: tangent plane $z = 5 + 2(x-1) + 4(y-2)$. Then $f(1.1, 1.9) \approx 5 + 0.2 - 0.4 = 4.8$; the exact value is 4.82.

For general n, $Df(\mathbf{a})$ is the row of partial derivatives. Its transpose is the *gradient* $\nabla f(\mathbf{a})$: next lecture.

Checking the definition

Example ($f(x,y) = xy$ is differentiable at every (a,b))

$f(a+h, b+k) = ab + bh + ak + hk$. With $L(h,k) = bh + ak$ the remainder is hk, and as $|hk| \le \frac{1}{2}(h^2+k^2)$: $\frac{|hk|}{\|(h,k)\|} \le \frac{1}{2}\|(h,k)\| \to 0$.

Example ($g(x,y) = \sqrt{|xy|}$ is not differentiable at $(0,0)$)

g is continuous, and $g_x(0,0) = g_y(0,0) = 0$, so the only candidate is $L = 0$. But along the diagonal: $\frac{g(t,t)-0}{\|(t,t)\|} = \frac{|t|}{\sqrt{2}\,|t|} = \frac{1}{\sqrt{2}} \not\to 0$.

A practical test

Theorem

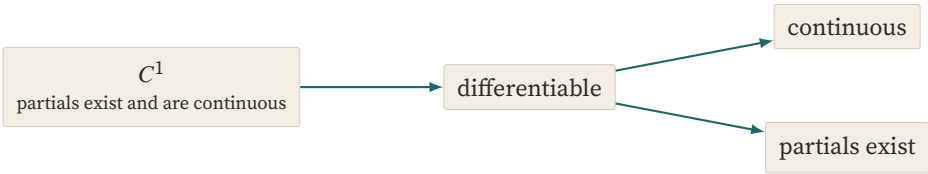
If all partial derivatives of f exist near $\mathbf{a}$ and are continuous at $\mathbf{a}$, then f is differentiable at $\mathbf{a}$.

Proof for $n = 2$.

Change one variable at a time, with the mean value theorem in each: $f(a+h, b+k) - f(a,b) = f_x(\xi, b+k)\,h + f_y(a,\eta)\,k$, with ξ between a and $a+h$, η between b and $b+k$. By continuity $f_x(\xi, b+k) = f_x(a,b) + o(1)$ and $f_y(a,\eta) = f_y(a,b) + o(1)$ as $(h,k) \to 0$. So the remainder is $o(1)\,h + o(1)\,k = o(\|(h,k)\|)$. □

Such functions are called C^1: polynomials, e^{xy}, $\sin(x^2 + y)$, $\|\mathbf{x}\|$ away from $\mathbf{0}$.

How the notions fit together



No arrow can be reversed:

- partials exist, not continuous: $\frac{xy}{x^2+y^2}$;
- continuous and partials exist, not differentiable: $\sqrt{|xy|}$;
- differentiable, not C^1: $(x^2+y^2)\sin\frac{1}{\sqrt{x^2+y^2}}$ at $(0,0)$ (practice).

In practice: compute the partial derivatives, check that they are continuous, and conclude.

04

The Jacobian matrix

The Jacobian matrix

For $f = (f_1, \dots, f_m)\colon \mathbb{R}^n \to \mathbb{R}^m$ the matrix of partial derivatives is the **Jacobian matrix**

$$J_f(\mathbf{a}) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & & \vdots \\ \frac{\partial f_m}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_n} \end{pmatrix}(\mathbf{a}), \qquad f(\mathbf{a}+\mathbf{h}) \approx f(\mathbf{a}) + J_f(\mathbf{a})\,\mathbf{h}.$$

Row i: the derivative of the component f_i. Column j: how f moves when x_j moves.

Example (Polar coordinates)

$P(r,\theta) = (r\cos\theta, r\sin\theta)$: $\quad J_P = \begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix}, \quad \det J_P = r.$

Why Jacobians matter

- **Linear approximation**: near $\mathbf{a}$, f acts like the matrix $J_f(\mathbf{a})$. Small changes $\mathbf{h}$ in the input become $J_f(\mathbf{a})\,\mathbf{h}$ in the output.
- **Chain rule** (Unit 12): $J_{g\circ f}(\mathbf{a}) = J_g\big(f(\mathbf{a})\big)\,J_f(\mathbf{a})$ — derivatives of compositions are products of matrices. Backpropagation in neural networks is this rule, applied layer by layer.
- **Change of variables** (Unit 20): $|\det J|$ measures how areas and volumes are stretched. For polar coordinates, $\mathrm{d}x\,\mathrm{d}y = r\,\mathrm{d}r\,\mathrm{d}\theta$.
- **Inverse function theorem** (Unit 17): if $\det J_f(\mathbf{a}) \neq 0$, then f can be inverted near $\mathbf{a}$.

05

Summary and practice

Summary

- **Partial derivative** $\frac{\partial f}{\partial x_i}$: differentiate in x_i, keep the other variables fixed. Mixed partials agree if they are continuous (Schwarz).
- Partial derivatives can exist where f is not even continuous.
- **Differentiable** at $\mathbf{a}$: $f(\mathbf{a}+\mathbf{h}) = f(\mathbf{a}) + Df(\mathbf{a})\,\mathbf{h} + o(\|\mathbf{h}\|)$ for a linear map $Df(\mathbf{a})$.
- Then f is continuous at $\mathbf{a}$, and $Df(\mathbf{a})$ is the **Jacobian matrix** of partial derivatives; for $m = 1$ it gives the tangent plane.
- Continuous partial derivatives (C^1) imply differentiability — the usual way to check it.

Practice

1. Compute all first partial derivatives: (a) $x^3y^2 - 4xy$ (b) $\ln(x^2+y^2+1)$ (c) x^y $(x>0)$ (d) $\frac{xyz}{x+y+z}$
2. Check that $f_{xy} = f_{yx}$ for $f(x,y) = e^{xy}\sin y$.
3. Find the tangent plane to $z = e^x \cos y$ at $(0,0)$, and use it to estimate $e^{0.1}\cos 0.2$. Compare with the true value.
4. Show from the definition that $f(x,y) = x^2 + y^2$ is differentiable at every point.
5. Is $f(x,y) = |x| + |y|$ differentiable at $(0,0)$?
6. Find the Jacobian matrix and its determinant for $F(x,y) = (x^2 - y^2,\ 2xy)$.
7. Let $f(x,y) = (x^2+y^2)\sin\frac{1}{\sqrt{x^2+y^2}}$ for $(x,y) \neq (0,0)$ and $f(0,0) = 0$. Show that f is differentiable at $(0,0)$ with $Df(0,0) = 0$.

Answers

1. (a) $f_x = 3x^2y^2 - 4y$, $f_y = 2x^3y - 4x$. (b) $\frac{2x}{x^2+y^2+1}$, $\frac{2y}{x^2+y^2+1}$. (c) $f_x = y\,x^{y-1}$, $f_y = x^y \ln x$. (d) with $s = x + y + z$: $f_x = \frac{yz(y+z)}{s^2}$, and similarly for f_y, f_z.
2. Both equal $e^{xy}\big((1+xy)\sin y + y\cos y\big)$.
3. $z = 1 + x$; the estimate is 1.1, the true value 1.0831. (The tangent plane ignores the second-order terms: Unit 13.)
4. $f(a+h, b+k) - f(a,b) - (2ah + 2bk) = h^2 + k^2 = \|(h,k)\|^2 = o(\|(h,k)\|)$.
5. No: $f_x(0,0)$ does not exist, since $x \mapsto |x|$ is not differentiable at 0.
6. $J_F = \begin{pmatrix} 2x & -2y \\ 2y & 2x \end{pmatrix}$, $\det J_F = 4(x^2 + y^2)$.
7. $|f(\mathbf{h}) - 0 - 0| \le \|\mathbf{h}\|^2$, so $\frac{|f(\mathbf{h})|}{\|\mathbf{h}\|} \le \|\mathbf{h}\| \to 0$. (Its partial derivatives are not continuous at $\mathbf{0}$, so it is not C^1.)

Next time

Lecture 11 — The gradient (Unit 11)

- Directional derivatives: the rate of change in any direction
- The gradient ∇f: direction of steepest ascent
- The gradient is perpendicular to level sets

Reading: Strang & Herman, *Calculus, Volume 3*, Section 4.6.

Questions: the module channel on Discord, or office hours, Monday–Friday, 10:00–17:00 (Baku time, UTC+4), ali.khudiyev@wosm.academy.